■ Explainable AI for Crop Disease Detection

College Minor Project Milestone 1 — Implementation Summary Report

Executive Overview & Status

This document provides a comprehensive technical reference of the Explainable AI (XAI) Crop Disease Detection module built in the workspace. It summarizes the project architecture, data acquisition, MobileNetV2 transfer learning pipeline, genuine Grad-CAM interpretability, Streamlit dashboard, and execution steps so you can copy/paste or reference this state in ChatGPT for further discussions.

Component	Implementation Status	Details
Project Structure	Completed	Modular architecture under crop-disease-xai/
Dataset	Downloaded Real Images	PlantVillage dataset (Tomato & Potato leaf disease categories)
Model Architecture	MobileNetV2 Transfer Learning	Pre-trained ImageNet backbone + GlobalAveragePooling + Softmax
Explainable AI	Genuine Grad-CAM	TensorFlow GradientTape targeting 'out_relu' feature layer
Web Application	Streamlit Dashboard	4 Tabs: Prediction, Explainability, Performance, About
Evaluation	Empirical Test Set	Accuracy, Precision, Recall, F1, Confusion Matrix figures
Testing & Docs	Verified	Pytest basic suite, README, VIVA.md, MID_EVALUATION.md

Project Files & Module Responsibilities

• **app.py**: Main Streamlit dashboard implementing 4 interactive tabs and custom CSS styling.

• **src/config.py**: Centralized configuration parameters, image sizes, paths, confidence thresholds, and palette.

• **src/dataset_downloader.py**: Automated fetcher downloading real PlantVillage images from GitHub.

• **src/data_preprocessing.py**: Image integrity validation, stratified train/val/test splits, and augmentation.

• **src/model.py**: MobileNetV2 backbone definition and dynamic Grad-CAM layer inspector.

• **src/train.py**: Two-phase transfer learning execution, ModelCheckpoint, EarlyStopping, and curve generation.

• **src/evaluate.py**: Test set evaluation, empirical metrics calculation, confusion matrix, and distribution plots.

• **src/gradcam.py**: Genuine TensorFlow GradientTape Grad-CAM heatmap computation and OpenCV overlay blending.

• **src/predict.py**: High-level inference pipeline returning structured prediction, top-3 probabilities, and Grad-CAM.

• **src/utils.py**: Qualitative confidence ratings, narrative explanations, and academic disclaimers.

• **tests/test_basic.py**: Unit tests for preprocessing, model output shapes, softmax sum, and Grad-CAM dimensions.

• **docs/VIVA.md**: 21 comprehensive viva voce questions and answers.

• **docs/MID_EVALUATION.md**: 12-slide mid-evaluation presentation outline.

Commands to Run the Project

1. Install requirements:

```
pip install -r requirements.txt
```

2. Download PlantVillage dataset images:

```
python src/dataset_downloader.py
```

3. Run unit test suite:

```
pytest tests/test_basic.py
```

4. Train MobileNetV2 model:

```
python src/train.py
```

5. Evaluate model on unseen test set:

```
python src/evaluate.py
```

6. Launch Streamlit XAI web dashboard:

```
streamlit run app.py
```

Future Scope Roadmap (Phases 2-4)

• **Phase 2**: Healthcare disease risk prediction using tabular ML + SHAP explanations.
• **Phase 3**: Food quality image classification + visual XAI.
• **Phase 4**: Unified Agriculture + Healthcare + Food Quality XAI platform.